

Proof-of-Concept: Simultaneous Acquisition of Real-Time MRI Video, EEG, and Surface EMG During Speech Production for Articulatory, Brain, and Muscle Activity Alignment

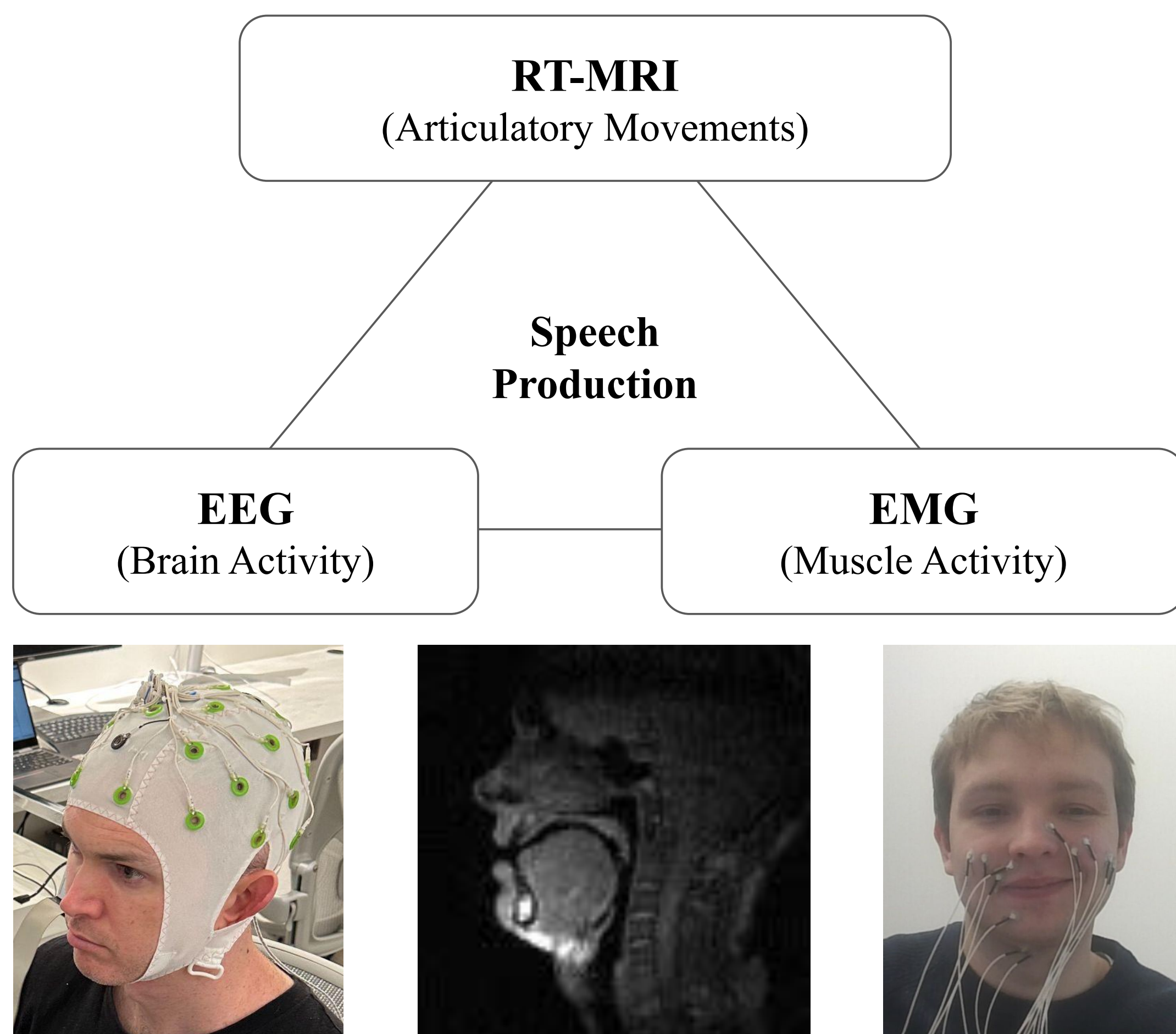
Jihwan Lee, Parsa Razmera, Kevin Huang, Sean Foley, et al.

USC Ming Hsieh Department of Electrical and Computer Engineering



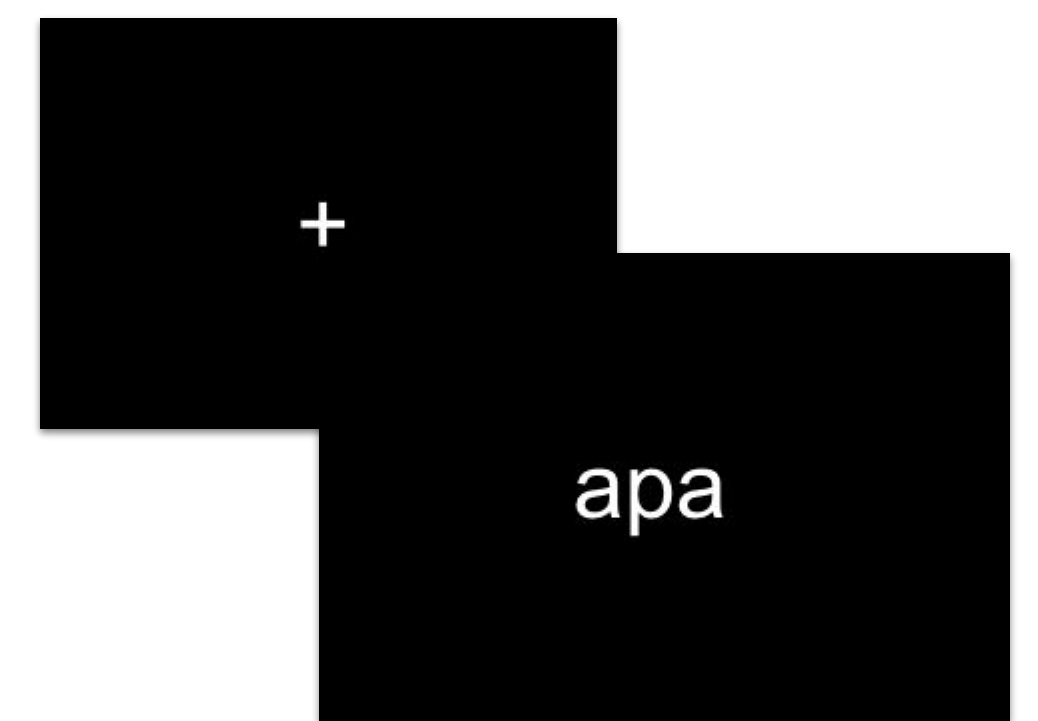
Introduction

- We propose the very first attempt for **parallel recordings** of the following biosignals during **speech production**: real-time magnetic resonance imaging (RT-MRI), electroencephalography (EEG), and electromyography (EMG).
- Aims to uncover the **neural** and **muscle connections** between **brain activity** and the mechanisms of **speech production**.
- How **neural** and **muscle signals** coordinate with **articulators** in the physical processes of **speech planning** and **production**.



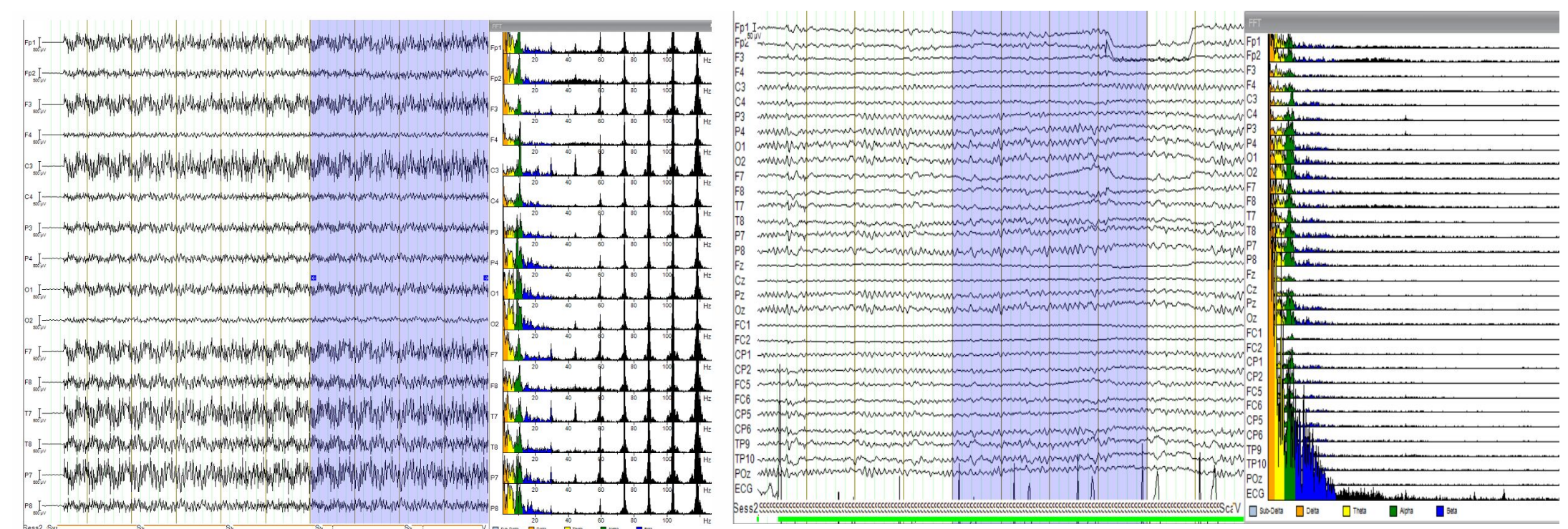
Methods

- Participant: One American English speaker.
- Speech types: **Phonated**, **attempted** (w/o vocalizing), and **imagined** speech production.
- Magnetic types: **inside** and **outside** the MRI scanner.
- Stimuli: 18 nonce words (aCa) (e.g., apa, aga, ...)
 - Random time break (black screen; 0.5, 0.75, 1 sec)
 - Eye fixation (+ in the center; 0.5 sec)
 - Target word appears (do not speak; 1 sec)
 - Target word disappears and start producing (2 secs)
 - Minimize visual interference in brain signals
 - Black screen (0.5 sec)

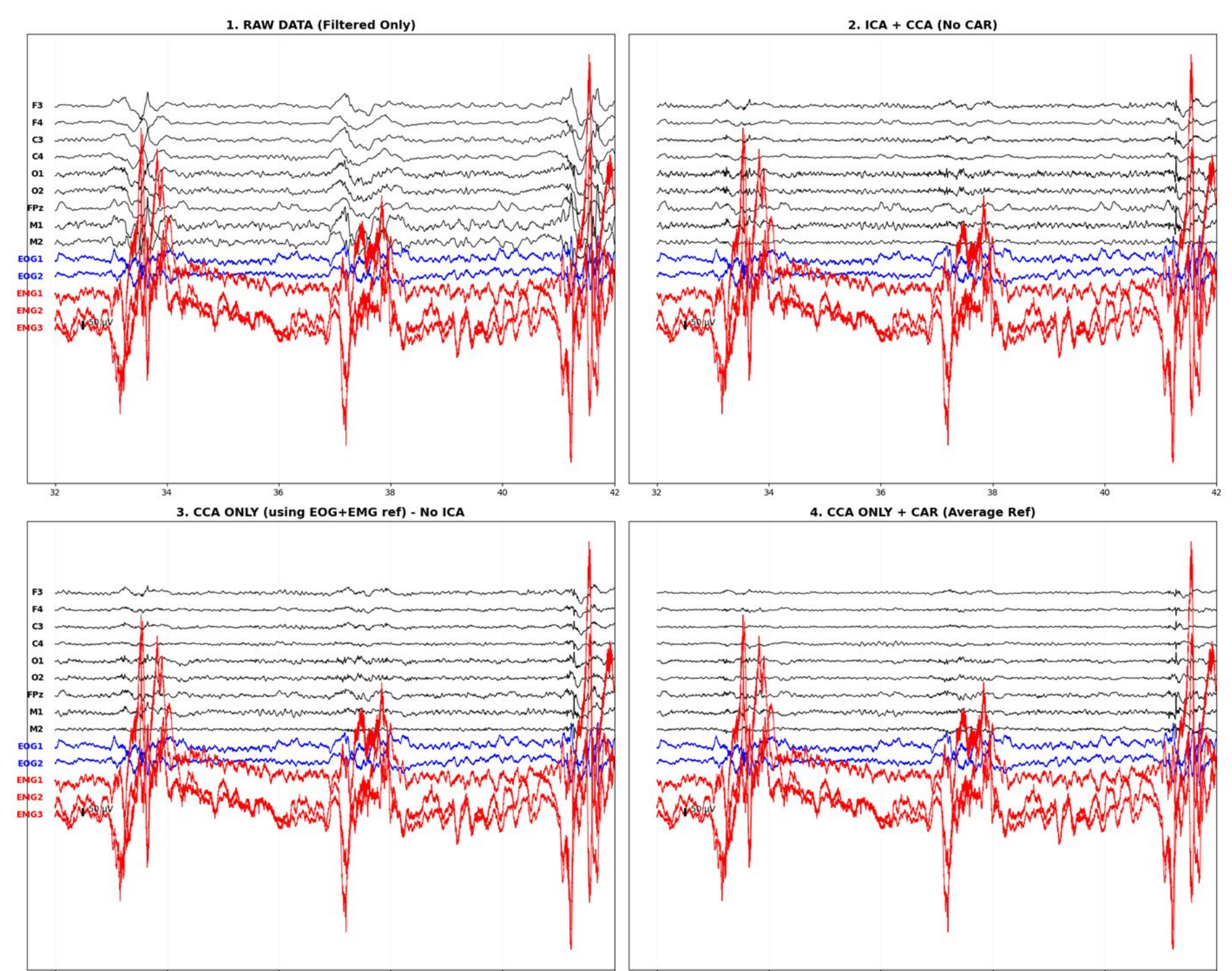


Preliminary Results

- Magnetic artifacts in EEG/EMG recordings



- Myopotential artifact correction in EEG recordings



Background

- **RT-MRI: temporal-dynamics** of **articulators** during **speech production**. We have been one of the pioneering research teams in this area.
- **EEG** is a non-invasive method to measure **brain signals**, revealing the relationship between the **neural signals** and **speech production**.
- **EMG** measures **muscle** activation and movement information, usually for 1) **muscle movement analysis** during speech production and 2) **correcting the muscle movement artifacts** in EEG recordings.
- Parallel recording of RT-MRI, EEG, and EMG during speech production aims to enhance our understanding of the **temporal-dynamics** between **neural signals** and the **movement** of the **articulators**.